

California Housing Dataset – Pandas, Visualization, and Feature Engineering Practice

Phase 1 – Basic Data Exploration (Pure Pandas)

- Load the California Housing dataset from sklearn and convert it to a pandas DataFrame.
- Add the target variable MedHouseVal as a column and rename columns to readable names.
- Inspect dataset shape, data types, and check for missing values.
- Compute summary statistics (mean, median, std, quantiles).
- Sort and filter districts by income, house age, and house value.
- Create derived columns such as rooms_per_household, bedrooms_per_room, population_per_household.

Phase 2 – Grouping and Aggregation

- Create income bins using pandas cut (e.g., 0–2, 2–4, 4–6, 6–8, 8+).
- Group by income bins and compute mean house values.
- Aggregate geographic statistics by latitude and longitude bins.
- Perform multi-level grouping (income_bin × house_age_bin).
- Use pivot tables to summarize statistics.

Phase 3 – Visualization Practice

- Plot histograms for house value, income, and house age.
- Compute and visualize the correlation matrix using a heatmap.
- Create scatter plots such as income vs house value and rooms_per_household vs house value.
- Plot longitude vs latitude and color points by house value.
- Experiment with marker size using population.

Phase 4 – Feature Engineering

- Create density-based features such as population_density and housing_density.
- Generate interaction features like income × house_age.
- Apply log transformations to skewed variables (income, population, house value).
- Detect outliers using IQR or z-score methods and visualize with boxplots.

Phase 5 – Advanced Pandas Skills

- Sort by latitude and compute rolling statistics such as rolling mean house value.
- Create ranking metrics for richest districts or most crowded districts.
- Construct pivot tables summarizing house values across bins.

Phase 6 – Analysis and Inference

- Rank features based on correlations with house value.
- Evaluate relationships visually and statistically.
- Test hypotheses such as whether higher-income districts have higher house values.
- Write a short analytical summary describing key patterns in the dataset.

Phase 7 – Challenge Exercises

- Recreate engineered features commonly used in housing prediction models.
- Identify 'luxury districts' defined as the top 5% house values.
- Build a reusable feature engineering function that takes a raw dataframe and returns engineered features.